

Interfacing memory and i/o ports

Memory

Memory

Store
Programs
and Data

Processor Memory

- They are tiny storage units inside the CPU.
- They hold data and results for a short time while the CPU works.
- They are very fast, so there's no speed difference between the CPU and registers — they work at the same speed.
- But because they are so fast and small, they make the CPU more expensive.

Primary or Main Memory

- Storage area which can be directly accessed by microprocessor
- Store programs and data prior to execution
- The memory should work at almost the same speed as the processor. For this reason, semiconductor memories are made using CMOS technology, which is fast and efficient.
- ROM, EPROM, Static RAM, DRAM

Secondary Memory

- Storage devices that are slower than RAM and registers.
- Examples: magnetic tapes, hard disks, CDs, DVDs, SSDs.
- Used to store large files and programs for a long time.
- Stores things like the operating system, compilers, databases, and other permanent files.
- Keeps data even when the computer is turned off.

Memory organization in 8086

- Memory ICs are byte-oriented, meaning each location stores 1 byte (8 bits).
- The 8086 is a 16-bit processor, so it handles data in words (2 bytes = 16 bits).
- A word is stored in two consecutive memory locations:

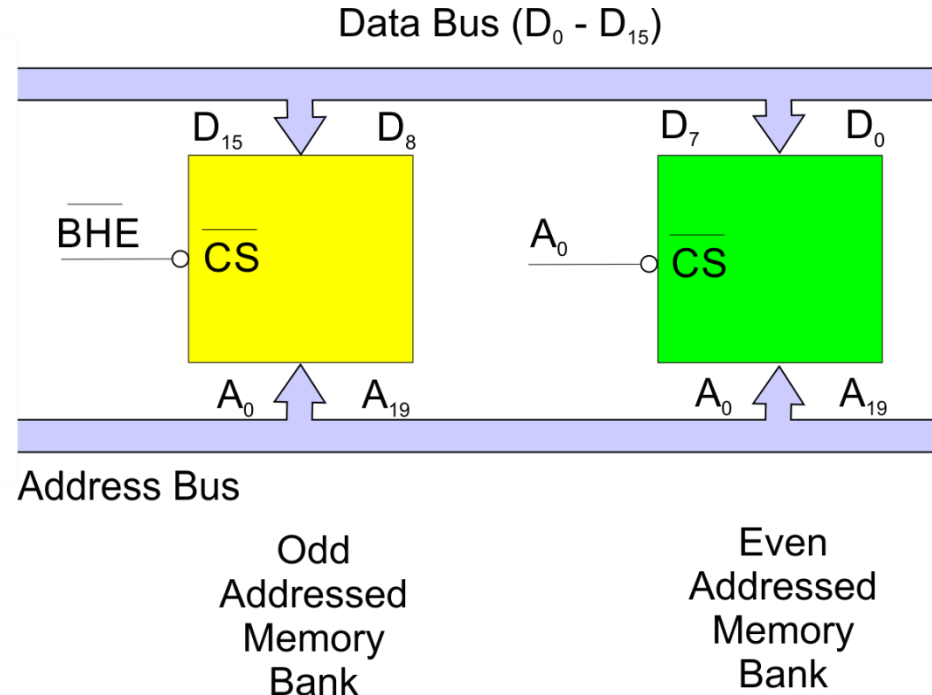
1. One location for the Least Significant Byte (LSB).

2. The next location for the Most Significant Byte (MSB).

- The address of a word is always defined as the address of its LSB. Memory is divided into two banks for efficient access:

1. Bank 0 (Even Addressed Bank): Selected when $A_0 = 0$, stores bytes at even addresses.

2. Bank 1 (Odd Addressed Bank): Selected when $A_0 = 1$, stores bytes at odd addresses.



What is BHE^- ?

BHE^- means Bus High Enable (active low).

It controls whether the higher byte (D15–D8) is used.

If $\text{BHE}^- = 0 \rightarrow$ High byte is active

If $\text{BHE}^- = 1 \rightarrow$ High byte is not active

What is A0 ?

A0 is the lowest bit of the address.

A0 = 0 $\rightarrow$ even address

A0 = 1 $\rightarrow$ odd address

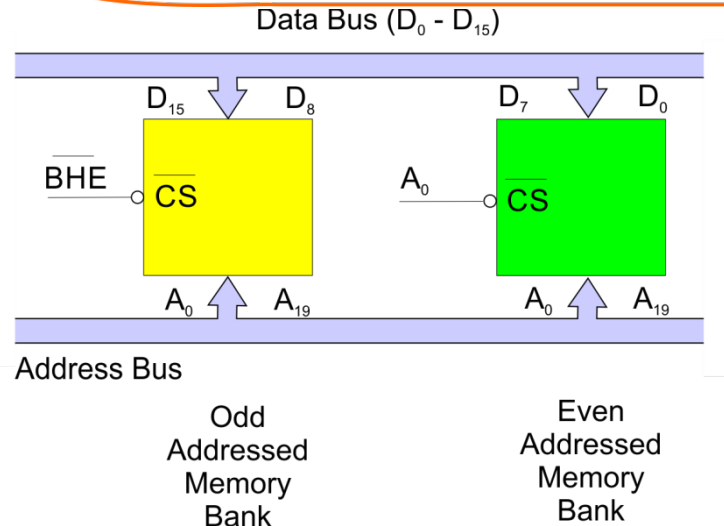
What are the data lines?

The 8086 microprocessor has 16 data lines: D0 – D15.
This means it can transfer 16 bits (2 bytes, i.e., one word) in a single operation.

D0–D7 = lower byte

D8–D15 = higher byte

Memory organization in 8086



	Operation	$\overline{BHE}$	A ₀	Data Lines Used
1	Read/ Write byte at an even address	1	0	D ₇ - D ₀
2	Read/ Write byte at an odd address	0	1	D ₁₅ - D ₈
3	Read/ Write word at an even address	0	0	D ₁₅ - D ₀
4	Read/ Write word at an odd address	0	1	D ₁₅ - D ₀ in first operation byte from odd bank is transferred
		1	0	D ₇ - D ₀ in first operation byte from even bank is transferred

The available memory space in an 8086 system is made up of EPROM and RAM.

This memory is divided equally between:

- 1) The even bank (for even addresses)
- 2) The odd bank (for odd addresses)

This is done because the 8086 works with 16 bits (2 bytes) at a time. To read or write a word, it needs:

- 1) One byte from the even bank
- 2) One byte from the odd bank

How is this implemented?

The total memory space (EPROM + RAM) is split equally between the two banks.

This ensures both banks can supply data simultaneously for 16-bit operations.

Implementation Options

1 Two ICs

One IC holds the even bank (connected to even addresses, $A_0 = 0$)

One IC holds the odd bank (connected to odd addresses, $A_0 = 1$)

2 Multiple ICs

For larger memory, each bank can use multiple ICs.

For example, several ICs can be connected in parallel to form the total even bank and odd bank capacity

Interfacing SRAM and EPROM

The memory interface connects the processor to semiconductor memory ICs and allows the processor to read from or write to the memory.

Memory Operations:

EPROM: Supports read operations only.

SRAM/RAM: Supports both read and write operations.

Requirements for Successful Memory Access:

1. Memory Access Time: Must be shorter than the processor's read/write cycle to avoid timing conflicts.

2. Control Signals:

Chip Select (CS): Activates the memory chip.

Read (RD) / Write (WR): Determines whether data is read from or written to memory.

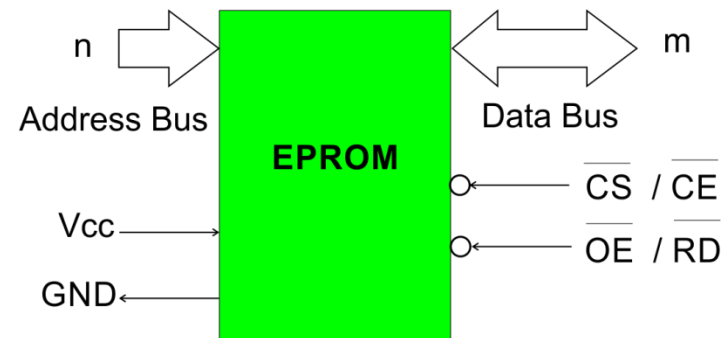
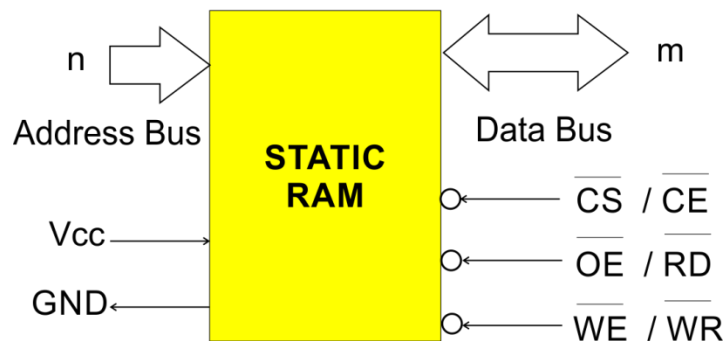
3. Addressing: Each memory location must have a unique address for proper access.

Key Notes:

Correct timing of control signals is essential for error-free operation.

SRAM allows fast read/write, while EPROM is read-only and slower to program.

■ Typical Semiconductor IC Chip



No of Address pins	Memory capacity			Range of address in hexa
	In Decimal	In kilo	In hexa	
20	$2^{20} = 10,48,576$	1024 k = 1M	100000	00000 to FFFFF

■ Memory map of 8086

The 8086 microprocessor can address up to 1 MB of memory, from 00000H to FFFFFH. The memory is divided into regions for RAM and EPROM.

RAM is mapped at the beginning (starting at 00000H) and is used for temporary storage, like variables, program data, stack, and the interrupt vector table (IVT). The IVT, which stores addresses for handling interrupts, is placed right at the start of RAM so the CPU can find it easily.

EPROM is mapped at the top end of the memory map (near FFFFFH) and holds permanent programs, such as monitor programs or BIOS. This ensures that when the CPU resets, it starts executing code from the EPROM automatically, allowing the system to boot and initialize correctly.

Monitor programs help set up devices like the keyboard/ display controller (8279) and other peripheral chips (8259, 8255, 8257) so the microprocessor can interact with hardware properly.

In short:

RAM → start of memory → temporary data, IVT, stack.

EPROM → end of memory → permanent startup code.

This setup ensures automatic startup and smooth hardware control.

- **EPROM (Erasable Programmable Read-Only Memory)** is a non-volatile memory that retains data without power. Data is programmed electrically using high voltage and can be erased with UV light. It is used in firmware, BIOS chips, and microcontrollers. EPROM is reprogrammable and retains data without power, but erasing is slow and requires special equipment, making it more expensive than regular ROM.

SRAM (Static Random-Access Memory) is a volatile memory that loses data when power is off. It stores bits using flip-flops, so no refresh is needed, making it very fast. SRAM is used in CPU caches, registers, and small memory buffers. It offers fast access and a simple interface but is costly and has lower storage density compared to DRAM.

A monitor program is a small software (firmware) stored in ROM that helps in controlling, testing, and debugging the system.

Monitor Programs

- ⇒ **Programming 8279 for keyboard scanning and display refreshing**
- ⇒ **Programming peripheral IC's 8259, 8257, 8255, 8251, 8254 etc**
- ⇒ **Initialization of stack**
- ⇒ **Display a message on display (output)**
- ⇒ **Initializing interrupt vector table**

Note :

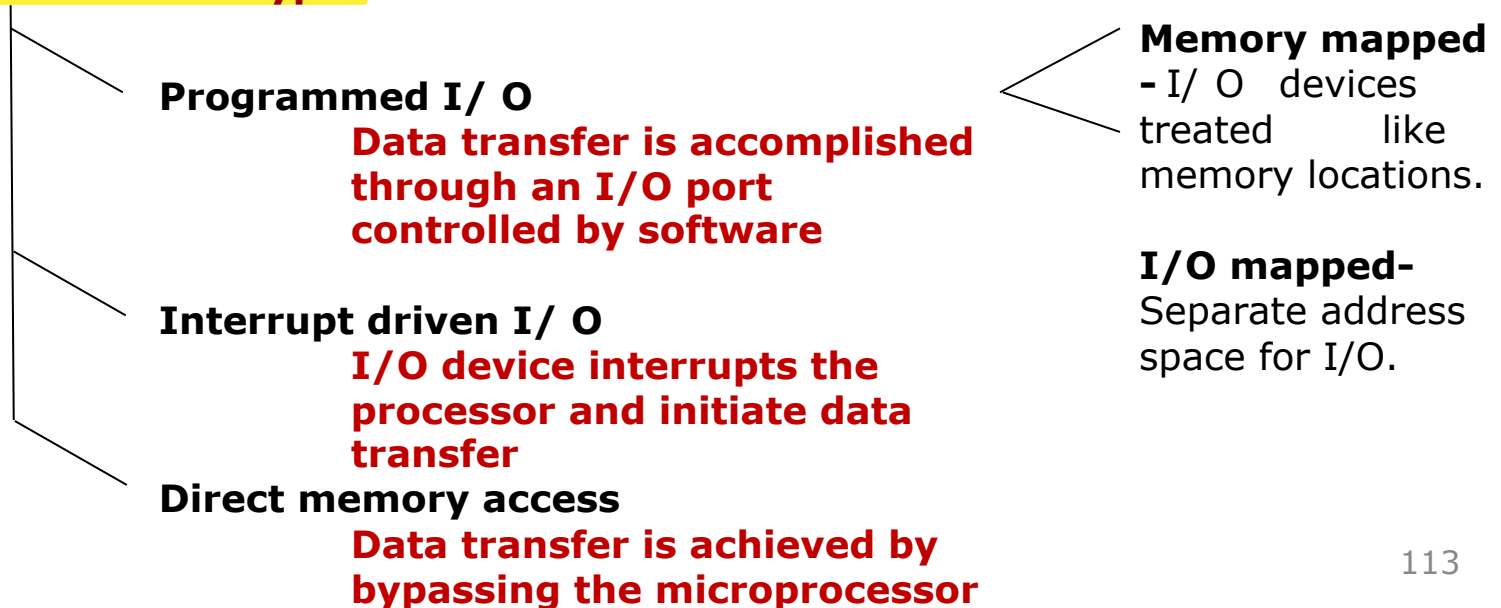
8255	Programmable peripheral interfacers
8257	DMA controller
8259	Programmable interrupt controller
8279	Programmable keyboard/ display controller

I/O devices

- ⇒ **For communication between microprocessor and outside world**
- ⇒ **Keyboards, CRT displays, Printers, Compact Discs etc.**



⇒ **Data transfer types**



8086 and 8088 comparison

Memory mapping	I/O mapping
20 bit address are provided for I/O devices	8-bit or 16-bit addresses are provided for I/O devices
The I/O ports or peripherals can be treated like memory locations and so all instructions related to memory can be used for data transmission between I/O device and processor	Only IN and OUT instructions can be used for data transfer between I/O device and processor
Data can be moved from any register to ports and vice versa	Data transfer takes place only between accumulator and ports
When memory mapping is used for I/O devices, full memory address space cannot be used for addressing memory.	Full memory space can be used for addressing memory.
⇒ Useful only for small systems where memory requirement is less	⇒ Suitable for systems which require large memory capacity
For accessing the memory mapped devices, the processor executes memory read or write cycle.	For accessing the I/O mapped devices, the processor executes I/O read or write cycle.
⇒ $M / \overline{IO}$ is asserted high	⇒ $M / \overline{IO}$ is asserted low